NATIONAL UNIVERSITY OF SINGAPORE
Department of Statistics and Data Science

ST5202 : Applied Regression Analysis

TUTORIAL 3 (Solution)

1. In a small-scale experiment study of the relation between degree of brand liking (Y) and moisture content (X_1) and sweetness (X_2) of the product. The data were obtained from the experiment based on a completely randomized design. The data is provided in the file `brand.txt` on Canvas.

   (i) The following multiple regression model is to be fitted to the data:

   $$\boldsymbol{y} = X\boldsymbol{\beta} + \epsilon.$$

   Give the explicit expressions of $\boldsymbol{y}, X$ and $\boldsymbol{\beta}$ using the data.

   The expressions are given below:

   $$\boldsymbol{y} = \begin{pmatrix} 64 \\ 73 \\ 61 \\ 76 \\ 72 \\ 80 \\ 71 \\ 83 \\ 83 \\ 89 \\ 86 \\ 93 \\ 88 \\ 95 \\ 94 \\ 100 \end{pmatrix}, \; X = \begin{pmatrix} 1 & 4 & 2 \\ 1 & 4 & 4 \\ 1 & 4 & 2 \\ 1 & 4 & 4 \\ 1 & 6 & 2 \\ 1 & 6 & 4 \\ 1 & 6 & 2 \\ 1 & 6 & 4 \\ 1 & 8 & 2 \\ 1 & 8 & 4 \\ 1 & 8 & 2 \\ 1 & 8 & 4 \\ 1 & 10 & 2 \\ 1 & 10 & 4 \\ 1 & 10 & 2 \\ 1 & 10 & 4 \end{pmatrix}, \; \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}.$$

   (ii) Write a short R program to fit the above regression model. Obtain the estimated regression function.

   The R program:

```
q1.dat = read.table("D://Rsession/brand.txt",header=TRUE)
y = q1.dat$degree; x1=q1.dat$moisture; x2=q1.dat$sweetness
q1.fit = lm(y~x1+x2)
summary(q1.fit)
```

The fitted results are as follows:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  37.6500     2.9961  12.566 1.20e-08 ***
x1            4.4250     0.3011  14.695 1.78e-09 ***
x2            4.3750     0.6733   6.498 2.01e-05 ***
Residual standard error: 2.693 on 13 degrees of freedom
Multiple R-squared:  0.9521,    Adjusted R-squared:  0.9447
F-statistic: 129.1 on 2 and 13 DF,  p-value: 2.658e-09
```

From the fitted results, the estimated regression function is obtained as

$$\hat{Y} = 37.65 + 4.425X_1 + 4.375X_2.$$

(iii) From the summary results obtained from the fitting, obtain the total sum of squares (SST) and the regression sum of squares (SSR). What is the proportion of the variation of Y which is explained by the two covariates?

From the fitted results, we have

$$F = MSR/MSE = 0.9521, \quad MSE = \hat{\sigma}^2 = 2.693^2 = 7.2522.$$

Thus,

$$\begin{aligned}
MSR &= F * MSE = 129.1 * 7.2522 = 936.2653, \\
SSR &= MSR * p = 936.2653 * 2 = 1872.531, \\
SSE &= MSE * (n - p - 1) = 7.2522 * 13 = 94.2786, \\
SST &= SSR + SSE = 1872.531 + 94.2786 = 1966.81.
\end{aligned}$$

The proportion of the variation of Y which is explained by the two covariates is $R^2 = 0.9521$.

(iv) Calculate the coefficient of determination $R^2_{Y,\hat{Y}}$ when Y is regressed on $\hat{Y}$ and the squared Pearson's correlation coefficient between Y and $\hat{Y}$. Are they the same? Do they equal to R^2?

To get $R^2_{Y,\hat{Y}}$, regress Y on $\hat{Y}$ using the following R code:

```
yhat = q1.fit$fitted
summary(lm(y~yhat))
```

The fitted results are as follows

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -9.948e-14  4.946e+00    0.00        1
yhat         1.000e+00  5.997e-02   16.67 1.25e-10
Residual standard error: 2.595 on 14 degrees of freedom
Multiple R-squared:  0.9521,    Adjusted R-squared:  0.9486
F-statistic:   278 on 1 and 14 DF,  p-value: 1.246e-10
```

$R^2_{Y,\hat{Y}}$ *is obtained from the summary results as 0.9521.*

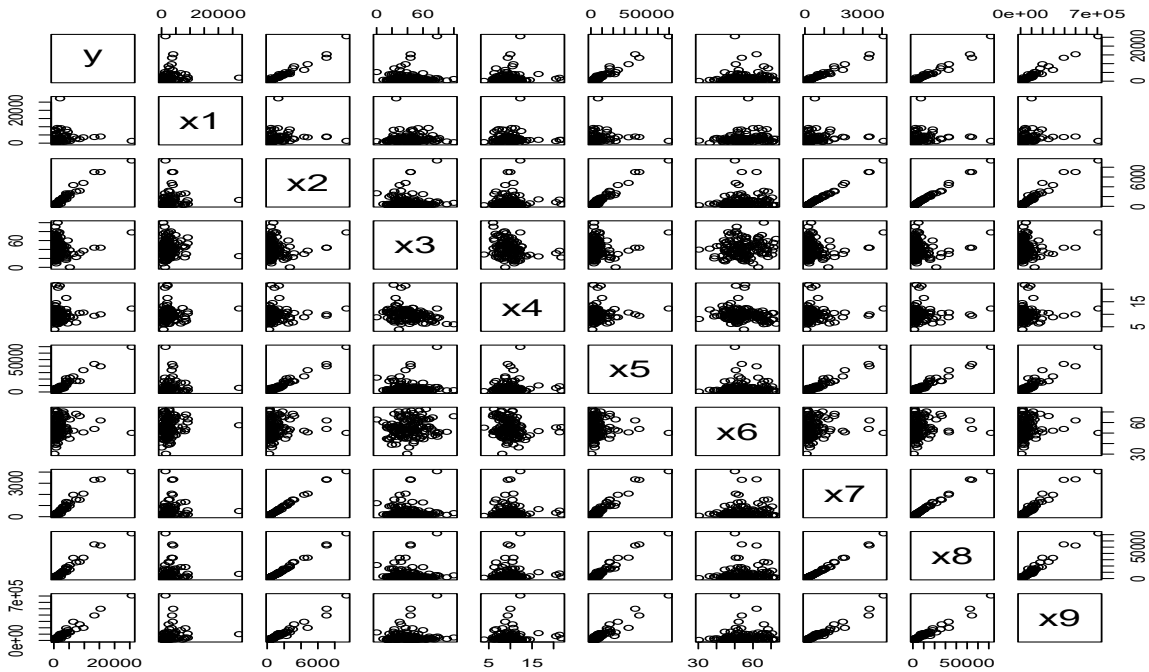
The squared Pearson's correlation coefficient between y and $\hat{Y}$ is computed by using `cor(y,yhat)^2`*, which yields the same value 0.9521. This value is the same as the value of R^2.*

2. Please refer to the SMSA data in Question 1, Tutorial 2.

   (i) Obtain scatter plots for each pair of the variables (except `geographic region`). Based on the scatter plots, propose a model for predicting the number of active physicians (y).

   The scatter plots below are obtained by the codes:

```
SMSA.dat = read.table("D://Teaching/22-23II5202/Rsession/SMSA.txt",
           header=TRUE)
attach(SMSA.dat)
y=doctor; x1=Land; x2=T.p; x3=P.city; x4=p.65; x5=beds; x6=h.sch;
x7=labor; x8 = income; x9=crimes
X=cbind(y,x1,x2,x3,x4,x5,x6,x7,x8,x9)
pairs(X)
```



From the plots, it can be seen that y has linear relations with the variables `x2`*,* `x5`*,* `x7`*,* `x8` *and* `x9`*. A linear regression model with these variables as covariates could be proposed.*

(ii) Consider the model for predicting the number of active physicians by using total population (x_2), civilian labor force (x_7) and total personal income (x_8) as predictor variables.

(iia) Fit the multiple regression model to the data. Obtain the estimated regression coefficients.

The fitted results are as follows:

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -71.47162   80.91374  -0.883  0.37862
x2            1.74061    0.62387   2.790  0.00602 **
x7           -5.65097    1.23172  -4.588 1.00e-05 ***
x8            0.40836    0.07294   5.599 1.14e-07 ***
Residual standard error: 580.7 on 137 degrees of freedom
Multiple R-squared:  0.9636,    Adjusted R-squared:  0.9628
F-statistic:  1210 on 3 and 137 DF,  p-value: < 2.2e-16
```

The estimated regression coefficients are given in the second column of the above table.

(iib) Compute the partial correlation coefficients between y and each of the predictor variables.

To obtain the partial correlation coefficients, first fit two regression models, one is that y regressed on the other variables, the other is that the concerned variable regressed on the other variables, then compute the Pearson's correlation coefficient of the two residuals from the fitted models. For the three predictor variables, the partial correlation coefficients are computed by the following codes:

```
fit11 = lm(y~x7+x8); fit12 = lm(x2~x7+x8);
ry1=fit11$resid; rx1=fit12$resid; cyx1=cor(rx1,ry1)

fit21 = lm(y~x2+x8); fit22 = lm(x7~x2+x8);
ry2=fit21$resid; rx2=fit22$resid; cyx2=cor(rx2,ry2)

fit31 = lm(y~x2+x7); fit32 = lm(x8~x2+x7);
ry3=fit31$resid; rx3=fit32$resid; cyx3=cor(rx3,ry3)
```

The partial correlation coefficients of y with x_2, x_7 and x_8 are respectively $0.2319, -0.3649$ *and* 0.4315.

(iic) Let $\tilde{\rho}_{yj}$ denote the partial correlation coefficient between y and x_j. Let σ_j denote the residual standard error when x_j is regressed on the remaining covariates and σ_{yj} the residual standard error when y is regressed on the remaining covariates. Verify numerically that

$$\hat{\beta}_j = \tilde{\rho}_{yj} * \sigma_{yj}/\sigma_j,$$

for each of the covariate in the model.

The verification is carried out by the codes:

```
sy1 = sqrt(var(ry1)); s1=sqrt(var(rx1)); cyx1*sy1/s1
sy2 = sqrt(var(ry2)); s2=sqrt(var(rx2)); cyx2*sy2/s2
sy3 = sqrt(var(ry3)); s3=sqrt(var(rx3)); cyx3*sy3/s3
```

The computed values are: $1.740609, -5.650972, 0.4083592$.

(iii) Let the number of serious crimes (x_9) be the response variable, and the population density (the total population, x_2, divided by land area, $x1$), total personal income (x_8) and percent high school graduates (x_6) be the predictor variables.

(iiia) Fit the model with the above response and predictor variables to the full data set and obtain the estimated regression function.

The R codes:

```
full.fit = lm(crimes~I(T.p/Land)+income+h.sch,data = SMSA.dat)
summary(full.fit)
```

The fitted results:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -9852.4294 11348.2760  -0.868    0.387
I(T.p/Land)   788.5265  1492.4394   0.528    0.598
income          8.6007     0.1799  47.810   <2e-16 ***
h.sch         179.7721   205.4787   0.875    0.383
```

(iiib) Fit the model in (iiia) for geographic region 1 only, obtain the estimated regression functions. Discuss its difference from the model fitted to the full data.

First extract the data from region 1 the codes:

```
SMSA1.dat = SMSA.dat[region==1],
```

The following codes fit the model for region 1:

```
R1.fit = lm(crimes~I(T.p/Land)+income+h.sch,data = SMSA1.dat)
```

The fitted results are as follows:

```
lm(formula = crimes ~ I(T.p/Land) + income + h.sch, data = SMSA1.dat)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2302.9645 55499.5148   0.041    0.967
I(T.p/Land)  2063.8882  2874.6678   0.718    0.480
income          8.9732     0.4498  19.951  5.1e-16 ***
h.sch        -376.8945  1047.2068  -0.360    0.722
```

Except that the effect of `income` *is significant in both fittings, the estimated regression functions are quite different. This implies that the region is an important factor which should be considered.*

3. The data taken on 20 incoming shipments of chemicals in drums arriving at a warehouse is provided in the file `ChemShip.txt` on Canvas. The data contains the number of drums in shipment (X_1), total weight of shipment (X_2, in hundred pounds), and number of minutes required to handle shipment (Y). A portion of the data is given below:

i :	1	2	$\cdots$	20
X_{i1} :	7	18	$\cdots$	11
X_{i2} :	5.11	16.72	$\cdots$	9.57
Y_i :	58	152	$\cdots$	90

(i) Fit the following regression model to the data:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i, i = 1, \ldots, 20.$$

Obtain the estimated regression function.

The model is fitted by the codes below:

```
q3.dat = read.table("D://Rsession/ChemShip.txt",header=TRUE)
y=q3.dat$time; x1=q3.dat$number; x2=q3.dat$weight
q3.fit=lm(time~number+weight,x=TRUE,data=q3.dat)
summary(q3.fit)
```

The Fitted results are:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   3.3243     3.1108   1.069      0.3
x1            3.7681     0.6142   6.135 1.10e-05
x2            5.0796     0.6655   7.632 6.89e-07
Residual standard error: 5.618 on 17 degrees of freedom
Multiple R-squared: 0.9869,  Adjusted R-squared: 0.9854
```

From the fitted results, the estimated regession function is obtained as

$$\hat{Y} = 3.3243 + 3.7681x_1 + 5.0796x_2.$$

(ii) Obtain the following quantities: (1) fitted values, (2) residuals, (3) the estimated regression coefficients, (4) the estimated covariance matrix of the estimated regression coefficients, (5) the multiple coefficient of determination R^2.

The multiple coefficient of determination R^2 is obtained from the summary results as 0.9869. The others are extracted from the fitted object `q3.fit` *by the following codes:*

```
yhat=q3.fit$fitted
r = q3.fit$resid
b = q3.fit$coef
V = vcov(q3.fit)
```

(iii) Compute the squared sample multiple correlation coefficient between Y and $Z = (\boldsymbol{x}_1, \boldsymbol{x}_2)$ by using the formula

$$\text{MCORR}(\boldsymbol{y}, Z)^2 = \frac{\boldsymbol{y}^\top (I - \frac{\mathbf{1}\mathbf{1}^\top}{n}) Z [Z^\top (I - \frac{\mathbf{1}\mathbf{1}^\top}{n}) Z]^{-1} Z^\top (I - \frac{\mathbf{1}\mathbf{1}^\top}{n}) \boldsymbol{y}}{\boldsymbol{y}^\top (I - \frac{\mathbf{1}\mathbf{1}^\top}{n}) \boldsymbol{y}}.$$

Note that $Z^\top (I - \frac{\mathbf{1}\mathbf{1}^\top}{n}) Z$, $Z^\top (I - \frac{\mathbf{1}\mathbf{1}^\top}{n}) \boldsymbol{y}$ and $\boldsymbol{y}^\top (I - \frac{\mathbf{1}\mathbf{1}^\top}{n}) \boldsymbol{y}$ are, respectively, the sample variance matrix of Z, sample covariance vector between Z and $\boldsymbol{y}$ and the sample variance of $\boldsymbol{y}$ upto a common factor $1/n$. They can be computed by the R function `var` and `cov`.

It is computed by the codes:

```
Z=cbind(x1,x2); v.z=var(Z)
v.zy=cov(Z,y);  v.y=var(y)
mcor2=t(v.zy)%*%solve(v.z)%*%v.zy/v.y
```

The value `mcor2` *= 0.9869259.*